

FAST FACTS

- Bone tissue is as strong as steel but as light as aluminum.
- More than half your bones are located in your hands and feet.
- Your first bones, when you were born, were made of flexible cartilage. This turned into bones as you grew older.

The skull shapes the head and face, surrounds and protects the brain, and houses the eyes and ears.

► **SMALLEST BONE** *The size of a rice grain, the stirrup (stapes) is the body's smallest bone. It is one of three bones, called ossicles, found inside each ear.*



The ulna is the inner bone of the forearm.

The radius is the outer forearm bone that forms the elbow joint with the ulna and humerus.

The humerus is the upper arm bone that links the shoulder to the elbow.

The breastbone (sternum) provides an attachment point for the ribs and protects the heart.

The ribs are 12 pairs of curved bones that support the chest and play a role in breathing.

The collarbone (clavicle) helps hold the shoulder and arm out to the side of the body.

The shoulder blade (scapula) forms the shoulder joint with the humerus.

The backbone is a flexible column of bones (vertebrae) that holds the head and upper body upright.

The hip girdle supports organs in the lower abdomen and forms a hip joint with each thighbone.



WILHELM ROENTGEN

In 1895, German physicist Wilhelm Roentgen discovered X-rays. He found that when X-rays were projected through the body onto a photographic plate, they produced clear images of bones.

SEEING THE SKELETON

Doctors use X-rays to see the skeleton inside living bodies. X-rays help doctors to diagnose damaged or broken bones. X-rays also reveal how some bones, such as the ribs, perform an important role by surrounding and protecting soft organs.

◀ **CHEST X-RAY** *This X-ray shows a man's healthy lungs (dark areas) and heart (pale blue) enclosed by the ribs (pale bands).*

